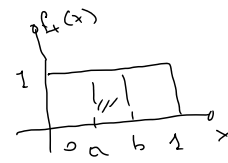
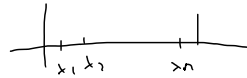


$$\{x_1, x_2, \dots, x_n\}$$

$$x_c \in (a, b)$$

$$\left\{ \frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n} \right\}$$

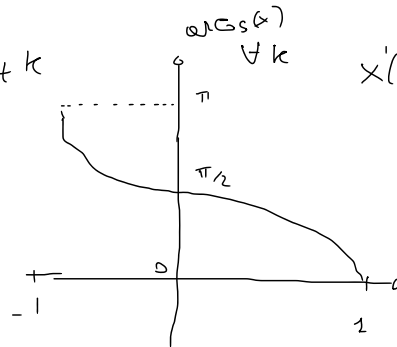
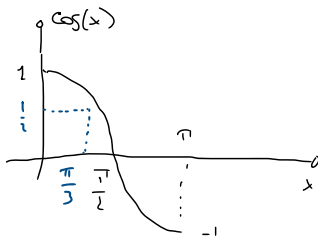


$$u_1, u_2, \dots, u_n$$

$$F_x^{-1}(u_1), F_x^{-1}(u_2), \dots, F_x^{-1}(u_n)$$

$$P[U \in (a, b)] = \int_a^b \pi(x - \frac{1}{2}) dx = b - a$$

$$x(y) = \left[\frac{1}{2} \arccos\left(\frac{y}{A}\right) - \frac{\pi}{2} \right] \frac{1}{2\pi} + k$$



$$x'(y) = \frac{1}{2\pi} \frac{(-1)}{\sqrt{1 - \left(\frac{y}{A}\right)^2}} \cdot \frac{1}{A}$$

$$|x'(y)| = \frac{1}{2\pi A \sqrt{1 - \left(\frac{y}{A}\right)^2}}$$

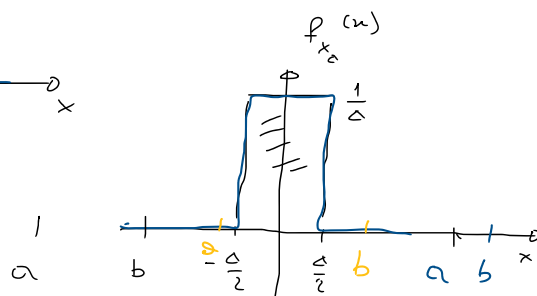
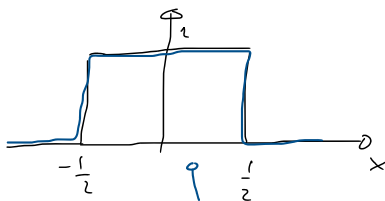
$$\cos[\arccos(x)] = x \quad x \in (-1, 1)$$

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$$

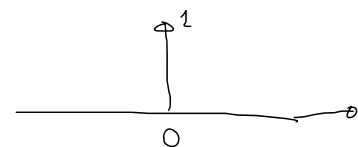
$$-\sin[\arccos(x)] \frac{d}{dx} \arccos(x) = 1$$

$$\frac{d}{dx} \arccos(x) = \frac{1}{-\sin(\arccos(x))} = \frac{1}{-\sqrt{1 - \cos^2(\arccos(x))}} = \frac{-1}{\sqrt{1 - x^2}} \quad |x| < 1$$

$$f_Y(y) = \sum_{k=-\infty}^{+\infty} f_X\left[x_k^{(+)}(y)\right] |x_k^{(+)}{}'(y)| + \sum_{k=-\infty}^{+\infty} f_X\left[x_k^{(-)}(y)\right] |x_k^{(-)}{}'(y)| = \frac{1}{2\pi A \sqrt{1 - \left(\frac{y}{A}\right)^2}} + \frac{1}{2\pi A \sqrt{1 - \left(\frac{y}{A}\right)^2}}$$



$$= \frac{1}{\pi A \sqrt{1 - \left(\frac{y}{A}\right)^2}}$$



$$\int_a^b f_{x_c}(u) du = \begin{cases} 0 & \text{se } 0 \notin (a, b) \\ 1 & \text{se } 0 \in (a, b) \end{cases}$$

$$\frac{g(0)}{\Delta}$$

$$\int_a^b g(x) f_{x_a}(x) dx = \begin{cases} 0 & \text{se } 0 \in (a, b) \\ \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} g(x) f_{x_a}(x) dx = \int_{-\frac{\Delta}{2}}^{\frac{\Delta}{2}} g(x) \frac{1}{\Delta} dx = g(0) & \Delta \rightarrow 0 \end{cases}$$

$-\frac{\Delta}{2}$ $\frac{\Delta}{2}$

$$g(x) f_{x_a}(x) \approx g(0) f_{x_a}(x)$$

$$\int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{(b^2 + ab + a^2)}{3}$$

$$\begin{aligned} \frac{b^2 + ab + a^2}{3} - \frac{1}{4} (a^2 + b^2 + 2ab) &= \left(\frac{1}{3} - \frac{1}{4} \right) a^2 + \left(\frac{1}{3} - \frac{1}{4} \right) b^2 + \left(\frac{1}{3} - \frac{1}{2} \right) ab \\ &= \frac{4-3}{12} a^2 + \frac{1}{12} b^2 + \frac{2-3}{6} ab \\ &= \frac{1}{12} (a^2 + b^2 - 2ab) = \frac{(b-a)^2}{12} \end{aligned}$$

$$f_X(x) = e^{-x} u(x)$$

$$Y = \begin{pmatrix} 1 \\ 2 \end{pmatrix} X$$

$$\begin{aligned} E[X^2] &= \int_0^{+\infty} x^2 e^{-x} dx = \left[-x^2 e^{-x} \right]_0^{+\infty} - \int_0^{+\infty} (2x) (-e^{-x}) dx \\ &= 2 \int_0^{+\infty} x e^{-x} dx = 2 \underbrace{E[X]}_{=1} = 2 \end{aligned}$$

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 = 2 - (1)^2 = 1$$

$$\sigma_Y^2 = \frac{1}{x^2} \quad E[Y^2] = \frac{2}{x^2}$$

$$Y = aX$$

$$g(x) = ax$$

$$g(x) = y$$

$$ax = y \xrightarrow{a \neq 0} x = \frac{y}{a}$$

$$\begin{cases} 0 \\ 1 \end{cases}$$

$$f_Y(y) = f_X(x_1(y)) |x_1'(y)| = \frac{1}{|a|} f_X\left(\frac{y}{a}\right)$$

$$\begin{aligned} x_1(y) &= \frac{y}{a} & x_1'(y) &= \frac{1}{a} \\ |x_1'(y)| &= \frac{1}{|a|} \end{aligned}$$

$$f_{aX}(x) = \frac{1}{|a|} f_X\left(\frac{x}{a}\right)$$

$$X = e^{-\lambda x} u(x)$$

$$x > 0 \quad \gamma = \frac{1}{\lambda} X$$

$$a = \frac{1}{\lambda}$$

$$f_Y(y) = \lambda e^{-\lambda y} \frac{u(\frac{1}{\lambda} y)}{u(y)}$$

$$\frac{y}{a} = \lambda y$$

$$Y = X + a$$

$$g(x) = x + a$$

$$g(x) = y$$

$$x + a = y \Leftrightarrow x = y - a$$

$$x(y) = y - a$$

$$x'(y) = 1$$

$$f_Y(y) = f_X(x(y)) \underbrace{|x'(y)|}_{=1} = f_X(y-a)$$

$$f_{X+a}(x) = f_X(x-a)$$